LEVEL UP ACADEMY

Class 9 | CBSE / GSEB | Science

Quick Revision: Tissues

Plant tissues

- **Meristematic** (dividing) tissue: apical (tips of root and shoot: length), lateral (cambium: girth), intercalary (base of leaves or internodes).
- **Simple permanent:** parenchyma (living, thin walls, stores food; chlorenchyma has chlorophyll; aerenchyma has air spaces for floating), collenchyma (thickened corners, flexibility, e.g. leaf stalk), sclerenchyma (dead, lignin-thickened, hard, e.g. husk of coconut).
- **Protective:** epidermis (single layer, waxy cuticle in dry areas, stomata for gas exchange and transpiration); cork (dead cells with suberin, in older stems).
- **Complex permanent:** xylem (tracheids, vessels, xylem parenchyma, xylem fibres) carries water and minerals; phloem (sieve tubes, companion cells, phloem fibres, phloem parenchyma) carries food. Only phloem fibres are dead in phloem.

Animal tissues

Tissue	Types and examples
Epithelial	Covers and protects: squamous (skin, lining of mouth), cuboidal (kidney tubules, glands), columnar (intestine), ciliated (respiratory tract)
Connective	Blood (plasma + RBC, WBC, platelets), bone, cartilage, ligament (bone to bone), tendon (muscle to bone), areolar, adipose (fat storage)
Muscular	Striated or voluntary (limbs); smooth or involuntary (stomach, intestine, iris); cardiac (heart; involuntary, contracts and relaxes rhythmically throughout life)
Nervous	Neurons: cell body, dendrites, axon; carry messages quickly

Remember: tendon = muscle to bone · ligament = bone to bone · cartilage smooths joint surfaces (also in the nose, ear, trachea)

Common mistakes

- Mixing up tendon and ligament.
- Saying xylem carries food. Xylem: water and minerals; phloem: food.

Quick check

1. Which tissue makes a plant grow in girth, and where is it found?

2. Name the muscle tissue in the heart, and give one way it differs from muscle in the arm.